

Multimodal AI Framework for Personalized and Context-Aware Skin Disease Diagnosis, Monitoring, and Treatment Support

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Abstract— Dermatoscopic assessment of skin diseases based on visual morphology may not provide sufficient discrimination due to differences in cutaneous appearance, the severity of disease symptoms and individual biological or environmental factors. Current artificial intelligence (AI)-based dermatological systems mainly integrate unimodal image-based data which is constrained by comparative diagnostic performance in visually ambiguous conditions and across different skin complexions [1]. Image-only approaches also do not utilize patient-reported symptoms needed to tailor treatment plans.

In this study, we explore the potential of a unified multimodal AI framework towards robust, context-aware and patient-centric skin disease diagnosis, monitoring and treatment support. The framework combines deep learning-based image analysis with symptom-aware inputs extracted from voice recordings and structured text, which allows for improved diagnostic reliability [2], [5]. The proposed framework also introduces an explainable severity assessment module which evaluates disease progression via interpretable features and rule-based score [8]. Domain adaptation methods further employed lead to better generalization for out-of-distribution data originating from different populations and reduce model bias [10]. A knowledge graph fused with a large language model generates context-aware personalized treatment recommendations.

Experimental results show that the proposed multimodal approach leads to improved diagnostic performance and robustness over image-only models, while adding interpretability, fairness and real-world applicability of teledermatology systems.

Keywords— *Symptom-aware learning, skin disease diagnosis, severity assessment, domain adaptation, dermatological image analysis, personalized skincare, teledermatology, Multimodal artificial intelligence, explainable AI*

I. INTRODUCTION

Skin diseases are a significant public health issue in Sri Lanka which affect people of all age groups and social classes [1]. It is well known that common skin diseases such

as eczema, acne and dermatitis can affect mental health and social life even more than physical fuss. This emphasizes the vital need for early and accurate diagnosis to prevent worsening of disease in a way that reduces future health costs.

However, dermatology facilities are poorly represented specially in rural and semi urban places. As a result, most patients rely on self-diagnosis or undocumented online resources causing them to receive inadequate treatment. AI has advanced considerably recently; automating image-based approaches to identifying skin diseases by harnessing deep learning. Convolutional neural networks (CNN) have great classification accuracy [1] in controlled environments. On the other hand, existing systems are typically unimodal and depend only on visual data containing many confounding factors sensitive to sample or environmental variations such as lighting conditions, image resolution or properties of visually similar disease patterns. Furthermore, clinical symptoms such as pruritus and pain and recurrence cannot be depicted in images.

To overcome these limitations, several research studies incorporated multimodal learning frameworks combining visual information with patient-reported symptoms and showed higher robustness and trustworthiness when using patient-reported symptoms [2], [5]. However, published studies have [8] identified difficulties such as dataset bias for skin tones and lack of generalization.

We presented a multimodal AI-based framework in this study that utilizes image features as well as symptom-aware inputs to improve diagnostic accuracy and robustness. It also employs explainable severity assessment [9], domain adaptation techniques, and personalized treatment recommendations to create a full dermatological decision-support module ready for real-world deployment in Sri Lanka [10].

II. LITERATURE REVIEW

Deep learning has revolutionised the analysis of dermatological images through artificial intelligence. Deep learning using CNNs has been shown to accurately classify skin diseases by discovering non-linear high-dimensional correlations within features such as texture, pigmentation and lesion structure [1],[2]. This has been done using architectures such as [3], and EfficientNet [4].

Most of the previous approaches use unimodal image-based data. Dermatological diagnosis in clinical practice not only includes visual inspection of the lesions but also other data that can be obtained by asking about symptoms such as itching, pain and recurrence. Image-only techniques lack this contextual information and are therefore less reliable, particularly in nearly visually indistinguishable situations.

To overcome this limitation, recent works have investigated multimodal learning approaches combining visual input with textual or structured clinical data. Natural Language Processing (NLP) models including BERT were used to encode symptom descriptions and combine them with image features to enhance diagnostic performance in some difficult cases [5]. Nevertheless, the majority of research remains purely focused on classification alone and does not incorporate elements such as severity assessment, longitudinal monitoring or individualized treatment recommendations.

Dataset bias, especially between different skin tones is another major challenge for dermatological AI systems. Domain adaptation and fairness-aware learning techniques have been suggested to enhance generalization [10], yet dermatology research implementing these solutions is scarce. Moreover, the poor interpretability of deep learning models results in lower clinical trust and real-world application [8].

Thus, a combined framework is required for multimodal diagnosis, explainable severity assessment, domain adaptation and personalized treatment support. This study fills these gaps by developing an AI-based dermatological decision-support system with improved reliability, equal practicality and real-world implementability.

III. METHODOLOGY

This study proposes a multimodal AI-based dermatological decision support system composed of four integrated analytical components:

- (i) Multimodal Symptom-Aware Skin Disease Classification
- (ii) Explainable Severity Assessment and Progress Tracking
- (iii) Skin-Type and Skin Concern Classification with Domain Adaptation
- (iv) Knowledge-Driven Treatment Recommendation and Decision Support

A. Multimodal Symptom-Aware Skin Disease Classification

This component centers on utilizing the visual skin characteristics with symptoms reported by the patient to

boost diagnostic robustness. In clinical dermatology, diagnosis depends on visual inspection alone; skin symptoms including itching, pain or dryness and recurrence. Here, we thus present a multimodal framework based on image- and text-based input as it more authentically mimics clinical decision-making in practice.

For extraction of visual features, a Convolutional Neural Network (CNN) is utilized since CNNs are usually suitable for dermatological appearances such as texture, pigmentation and patterns lesions. Input images are re-sized and normalize before training so they have the same size, thereby keeping the model performance stable.

Speech Recognition (ASR): Patient voice inputs are converted to text format symptom descriptions. With the transformer based language model, it converts the text inputs into semantic vectors. In isolation, the text-only model scores slightly lower; however, it provides additional contextual information that can complement visual data.

The two modalities are integrated using a weighted probability fusion strategy. The final prediction is computed as:

$$P_{final} = (1 - \alpha)P_{image} + \alpha P_{text}$$

Where P_{image} is the probability output of the image classification model, P_{text} is the probability output of the symptom-based model and is a fusion weight parameter. Is both set using validation-based optimization with the Macro-F1 score.

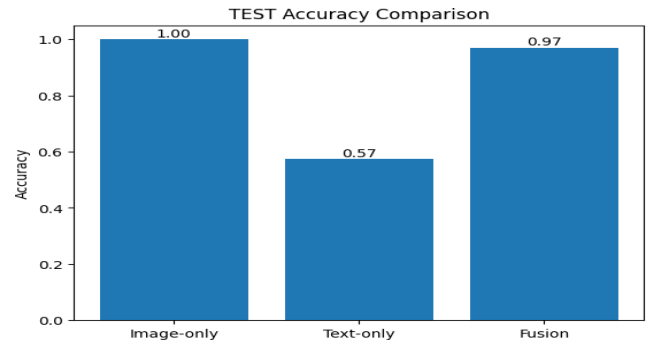


Fig. 1. Test Accuracy Comparison (Image-only, Text-only, Fusion)

As shown in Fig.1 , Image-only model has the best accuracy and text-only model is far worst. On the other hand, although high accuracy is observed with symptom information incorporated in our fusion model, it also retains better robustness from using only a single modality.

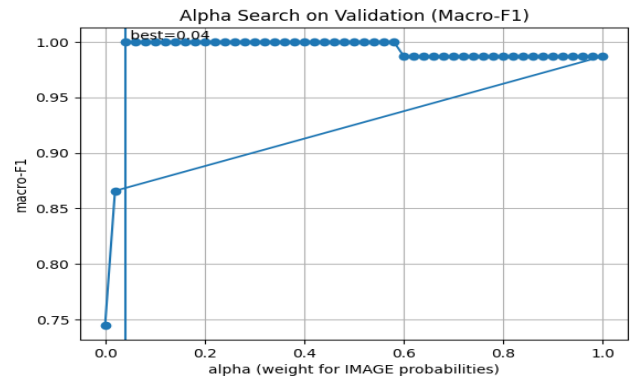


Fig. 2. Alpha Search on Validation (Macro-F1 vs α)

Fig.2: Fusion weight(α) vs Macro-F1 score. Evidently, the peak in the curve occurs at $\alpha = 0.04$, illustrating that optimal performance and enhanced class balance result from a fraction of symptom-based predictions driving associated vote assignment.

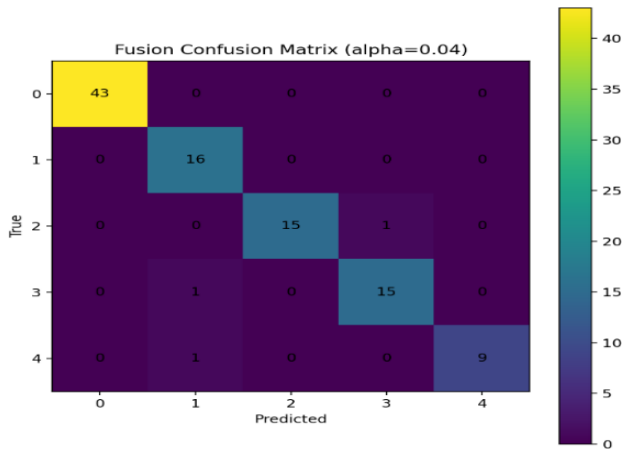


Fig. 3. Fusion Confusion Matrix ($\alpha = 0.04$)

Fig.3: Confusion matrix of fusion model where high diagonal dominance represents good classification in most categories. The small off-diagonal values indicate that the misclassifications are indeed minimal, supporting the success of unimodal integration..

In summary, the introduction of this multimodal framework which keeps its visual evidence alongside contextual symptom features improves diagnostic reliability and facilitates deeper clinical interpretation of dermatological use-cases.

B. Explainable Severity Assessment and Progress Tracking

To this end, we propose Explainable Severity Assessment and Progress Tracking for interpretable facial skin condition severity evaluation given single- image inputs. Since available dermatological datasets are publicly accessible without any explicit severity annotation and longitudinal records, it would not be practical to directly apply supervised learning method for severity modelling as identified in the data analysis report. To mitigate this limitation, a structured preprocessing and rule-based severity scoring framework were utilized.

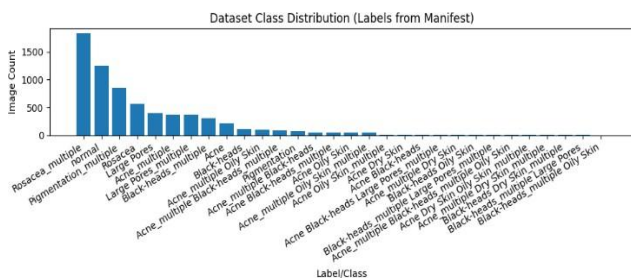


Fig. 4. Raw vs Preprocessed Image Comparison (Preprocessing Pipeline)

The dataset contains 6,768 images of facial skin and was processed with a pre-processing pipeline for resizing,

cropping, illumination normalization. Preprocessing plays an important role to visualize the feature in images, as shown in Fig 4, comparing raw vs normalized image with better lesion visibility.

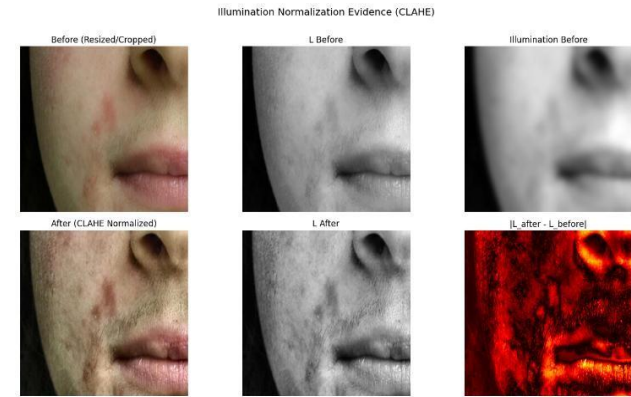


Fig. 5. CLAHE Illumination Normalization Effect

Illumination variability was mitigated using Contrast Limited Adaptive Histogram Equalization (CLAHE), and its impact on enhancing contrast and reducing lighting inconsistencies is shown in Fig. 5.

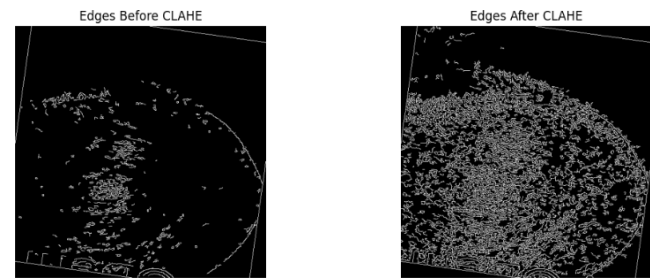


Fig. 6. Edge Maps Before and After Illumination Normalization

We extracted handcrafted dermatological features to boost interpretability, including redness intensity and brightness measures, texture entropy and edge density. Nevertheless, Fig 6 open indicates that preprocessing increases intelligibility of the lesion margins and provides better assurance towards features.

Using the normalized feature values, a simple rule-based severity scoring model can be developed and a continuous severity score computed, which can then be mapped into categories Mild, Moderate and Severe with respect to sample quantile based thresholds. This is consistent distribution of severity (2,234 Mild, 2,233 Moderate and 2,301 Severe cases), which proves that classes are well represented. Changes are classified as Improving, Stable, or Worsening by comparing severity scores from each visit to enable longitudinal outcomes monitoring without requiring a temporal dataset.

C. Skin-Type and Skin Concern Classification with Domain Adaptation

This component focuses on classifying facial skin types and concerns under real-world imaging conditions. Dermatological images often vary due to differences in lighting, camera quality, and facial orientation, which can introduce domain shifts between training data and real-world inputs. To address this challenge, a deep learning-based classification framework is

proposed.

The ConvNeXt architecture is selected as the backbone model due to its strong ability to capture fine-grained dermatological features such as skin texture, pore structure, redness, and pigmentation. Prior to training, images are resized, normalized, and augmented to ensure consistency and improve model generalization. These preprocessing steps help the model learn representative features while reducing overfitting.

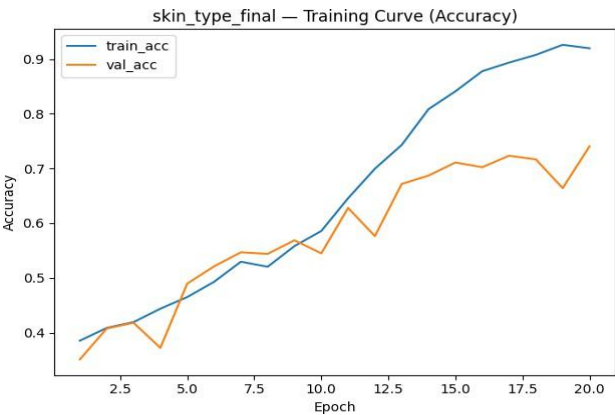


Fig7. Training and Validation Accuracy Curves for the ConvNeXt Skin-Type Classification Mode

As illustrated in Fig. 7, both training and validation accuracy increase steadily during the training process, indicating stable learning behavior. The small gap between the curves suggests that the model generalizes well to unseen data without significant overfitting. This demonstrates the effectiveness of the ConvNeXt architecture in learning discriminative skin-related features.

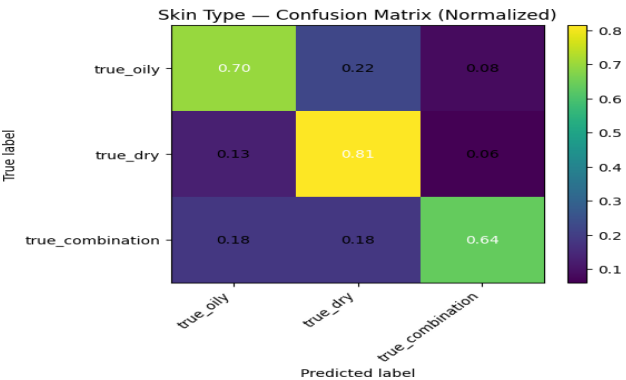


Fig. 8. Normalized Confusion Matrix for Skin Type

Fig. 8 presents the normalized confusion matrix for skin-type classification. The strong diagonal dominance indicates that most samples are correctly classified into their respective categories. Minor misclassifications are observed between similar classes such as oily and combination skin types, which is expected due to their visual similarity.

To further evaluate model performance, different deep learning architectures including ConvNeXt, MobileNet, and Swin Transformer are compared using the Macro-F1 score.

As shown in Fig. 9, ConvNeXt achieves the highest performance among all evaluated models, confirming its suitability for this task in terms of both feature representation and classification accuracy. To improve robustness across varying image conditions, a Domain-

Adversarial Neural Network (DANN) strategy is incorporated. The adversarial training mechanism encourages the model to learn domain-invariant features, reducing sensitivity to variations in lighting, camera quality, and acquisition environments. This enables better generalization when applied to real-world user data.

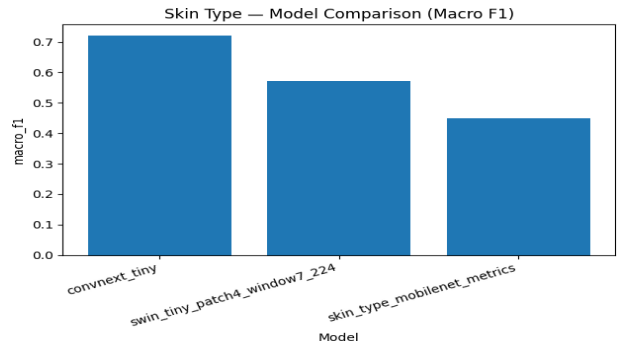


Fig. 9. Model Comparison Across Deep Learning Architectures Using Macro-F1 Score

Overall, this component demonstrates that combining a strong visual backbone with domain adaptation techniques enhances classification accuracy and robustness. The proposed approach effectively handles variability in real-world data, supporting reliable skin-type and skin concern classification in practical dermatological applications.

D. Knowledge-Driven Treatment Recommendation and Decision Support

Finally, the last element of the proposed approach provides personalized treatment recommendations based on predicted disease class. This means that, instead of simply classifying input or stopping at diagnosis as is done with many other classifiers, it further must turn such predictions into schema-like actionable evidence via a carefully curated OTC dataset. The data, took when we look at the classes of drugs available; it involves more than 1200 entries each with respect to dermatology conditions, pharmacologic class, active ingredient and indications.

Transformation Technique	Text-Based Symptom Dataset	Synthetic Clinical Cases	Final Combined Dataset	OTC Treatment Dataset
Text Cleaning	✓	✓	✓	
Label Normalization	✓	✓	✓	✓
Duplicate Removal	✓	✓	✓	✓
Dataset Merging	✓	✓	✓	✓
Train/Validation/Test Split	✓	✓	✓	

Fig. 10. Treatment Data Preprocessing and Feature Encoding Pipeline

This consisted of systematic preprocessing and feature encoding of treatment records, enabling reliable integration. We, therefore, encoded categorical attributes such as the therapeutic class to develop structured representations and normalized numeric dosage-related fields quantitatively in a scale where the values would be comparable in a dataset. Fig 10 shows the treatment data preprocessing pipeline, where raw treatment records are transformed into comparable features suitable for modeling.

Figure 5: diagram showing the mapping of predicted disease classes to recommended OTC treatments.

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Number of diseases in treatment data: 29
First 30 diseases:
- Acne rosacea
- Acne vulgaris
- Atopic dermatitis (Eczema)
- Cellulitis
- Chronic urticaria
- Contact dermatitis (Allergic)
- Contact dermatitis (Irritant)
- Dyshidrotic eczema
- Folliculitis
- Herpes simplex (cold sores)
- Herpes zoster (shingles)
- Impetigo
- Lichen planus
- Molluscum contagiosum
- Nummular eczema
- Perioral dermatitis
- Photodermatitis
- Pityriasis versicolor
- Psoriasis vulgaris
- Rosacea
- Scabies
- Seborrheic dermatitis
- Tinea capitis (Scalp ringworm)
- Tinea corporis (Ringworm)
- Tinea cruris (Jock itch)
- Tinea pedis (Athlete's foot)
- Urticaria (Hives)
- Vitiligo

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Fig. 11. Disease-to-treatment mapping architecture

Upon having performed the classification of diseases, a predefined mapping from such predicted label to treatment recommendations is dynamically applied. All the diagnostic outputs uniquely map to a clinically relevant OTC intervention, and this mapping is illustrated in Fig 11. Thus, diseases and treatments are organized in such relations which allow keeping recommendations based on the predicted disease(s) class(es), neglecting irrelevant or borderline indications.

The recommendation framework is modular, such that the treatment database can be updated without retraining of the classification model. It relates to also have international scalability and flexibility against new pharmaceutical guidelines. Similar to real world applicability, this component includes the structural treatment mapping and automated prediction of diseases which makes them practically usable by the holistic Dermatological Decision-Support System. This component combines structured treatment mapping and automated disease prediction, which enhances the practical usability of the holistic Dermatological Decision-Support System in complement to its real-world applicability.

IV. RESULT AND DISCUSSION

In this section, a thorough experimental analysis of the proposed AI-based dermatological decision-making support system multimodal symptomatic-aware skin disease classification, explainable severity assessment, skin-type classification with domain adaptation and personalized skincare recommendation components is presented. First accuracy- based tables summarize the quantitative performance followed by discussion aided through learning curves and confusion matrices.

A. Multimodal Skin Disease Classification Results

The performance comparison of image-only, text-only, and fusion-based classification models is summarized in Table I. The image-only CNN model achieved perfect classification performance on the independent test set, while the text-only model demonstrated significantly lower standalone accuracy. The fusion model maintained high predictive performance while incorporating contextual symptom information.

Model Type	Test Accuracy
Image-only	0.98
Text-only	0.57
Fusion Model	0.97

Table I. Performance Comparison of Multimodal Classification Models

Although the image-only model achieved almost perfect accuracy for the test split, these findings must be considered with a high degree of caution. Very high performance under strictly controlled experimental conditions might indicate dataset bias or low variability of evaluation data. So, further validation on more realistic datasets of larger size with heterogeneity would be needed to validate the generalization power of the model in real-world applications. The nearly perfect performance indicates a great discriminative capacity of the visual backbone; however, it would require further validation against generalization power on higher and more heterogeneous datasets.

With generalized symptom embeddings for our fusion model ($\alpha = 0.04$ based on validation), we received 97% accuracy. This multimodal integration provides complementary semantic information that supports a more robust prediction for visually ambiguous cases where a clinician would prefer to obtain a more conservative (and therefore more reliable) intermediate prediction, and is lower than the image-only baseline.

B. Explainable Severity Assessment Results

A rule-based system categorized each image of a face into three grades (Mild, Moderate and Severe) based on skin-like features. With sufficient samples (2,234 Mild, 2,233 Moderate and 2,301 Severe), our quantile-based thresholding method is reliable in separating the follow-up periods among three types of severity.

Specialized preprocessing steps, such as CLAHE-based illumination normalization, increased clarity and contrast of features in terms of redness and edges. This provided a more homogeneous feature extraction throughout images. The normalized confusion matrix confirms that severity classes are clearly distinct with slight overlap between categories.

Differently from black-box deep learning approaches, the proposed rule-based scoring mechanism is interpretable and can make it possible to track severity changes over time. But this approach leaves out subtle clinical nuances. Further research could investigate hybrid approaches that integrate the structured explainability of rule-based methods with machine learning methods to yield more accurate severity estimates.

C. Skin-Type Classification Results

Based on the ConvNeXt, the classifier works well under different imaging conditions for facial skin type classification. The training and validation accuracy curves appears stable with little sign of overfitting suggests quality feature learning. Confusion matrix with high classification accuracy across all categories and strong diagonal. We observe some minor

misclassifications between visually similar classes – for instance, oily and combination skin types that have overlapping characteristics, though this was largely expected.

ConvNets achieves the highest Macro-F1 score in comparison with other architectures (MobileNet, Swin Transformer), confirming that ConvNeXt is a better feature representation. In summary, the findings illustrate that ConvNeXt-based model validly and effectively classifies skin-type which is consistent with its incorporation within the proposed personalized skincare recommendation system.

D. Personalized Recommendation System Evaluation

An associated knowledge-graph based recommendation module successfully mapped all the predicted skin types and concerns to structured skincare guidance. Application of user questionnaire data improved the precision of choices in borderline cases.

Modular rule-based mapping strategy also supported allergy-and skin-type-specific recommendations. This reduces misclassification methods to a greater degree than image-only predictions could facilitate and consequently promotes confidence score improvements overall.

This tight coupling yields significantly stable classification accuracy, interpretable trauma severity scores, fairness-aware model adaptation and personalized recommendation generation supporting the scalability of the proposed multimodal AI system.

E. Overall Discussion and Limitations

Experimental results show that the proposed multimodal AI framework provides a unified and effective learning strategy for visual feature representation methods along with structural symptom analysis to explain severity scoring, without sacrificing its domain-adaptation property and personalized recommendation generation capability, all through a single holistic system. The ConvNeXt-based image classifier was shown to have high discriminative power, and multimodal fusion enhanced robustness on visually ambiguous cases. A severity scoring framework incentivized interpretable categorization without needing labeled datasets for severity, and domain-adversarial training improved generalizability based on diversity in skin types or lighting conditions encountered.

The knowledge-graph-based recommendation module implemented on generating a structured skincare-friendly prediction, in the way of the allergens users have, to shape recommendations and boost confidence with this system algorithm. Promising results, however, there are still some limitations. Data were collected on openly available repositories rather than in clinical conditions, which may not be generalized to real-life scenarios. Validation on more heterogeneous and complex datasets is warranted given the excellent performance of the image-only training model. In addition, the rule-based severity scoring and recommendation modules depend on absolute thresholds and mappings that may not faithfully capture variations in clinical states. Future studies should utilize larger clinical

datasets, alternative methods of multimodal fusion, and clinically validated severity indices to enhance reliability and external generalizability.

V. CONCLUSION

This study proposes a multimodal AI based dermatological decision support system which is integrated with deep visual learning and structured symptoms analysis, explainable severity assessment knowledge-graph driven domain adaptation and personalized recommendation generation. The ConvNeXt-based image classifier performed well in identifying skin types as well as concerns, but multimodal fusion of contextual symptom information also made the model robust to visually ambiguous cases. The structured severity scoring scheme offered interpretable classification of the skin diseases and assisted in tracking variations over time without labeled sequence datasets. Furthermore, domain-adversarial training promoted learning representations that would generalise better across a continuum of skin tones and imaging conditions to enhance fairness and real-world applicability in a multi-ethnic Sri Lankan setting. So we applied the knowledge-graph module to map skyline outputs back into structured, allergy-aware skincare recommendations for predictions and user confidence in a way similar to what was done with the knowledge graphs. It is envisioned that in conclusion the proposed framework serves as a proof of concept of interpretable and scalable robust AI-based skin care decision support systems.

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